

Thermal Monitor

Embedded-style thermal monitor implemented in C and C++.

Full project documentation is in docs/README.md.

Architecture diagram: docs/architecture.svg

Generated PDF documentation: docs/README.pdf

QEMU VM setup: qemu/README.md

QEMU integration tests require an x86_64 QEMU binary. On Debian/Ubuntu, install qemu-system-x86, which provides qemu-system-x86_64.

Prerequisites

On Debian/Ubuntu:

```
sudo apt update
sudo apt install git ca-certificates build-essential cmake libgtest-dev qtbase5-dev qt5-qmake
```

For the QEMU hardware integration test:

```
sudo apt install qemu-system-x86 qemu-utils cloud-image-utils openssh-client rsync
```

Quick Start

```
cmake -S . -B build
cmake --build build
ctest --test-dir build --output-on-failure
```

The build includes the C implementation, the C++ implementation, Qt simulator, mocks, unit tests, and real-Linux/QEMU demo targets.